

PROJECT REPORT

Automatic Street Light Using Arduino and LDR

Wokwi Simulation-Based Embedded Systems Project

1. Abstract

This project presents an automatic street light control system using an Arduino Uno and an LDR (Light Dependent Resistor). The LDR senses surrounding light intensity and provides an analog signal to Arduino analog pin A0. The Arduino reads the sensor value, compares it with a predefined threshold, and controls an LED connected to digital pin D8. When the simulated environment becomes dark, the LED is switched ON; when sufficient light is detected, it is switched OFF. The complete project is designed and tested using the Wokwi online simulator.

2. Introduction

Automatic lighting can reduce unnecessary energy consumption by operating lights according to actual environmental conditions. This project demonstrates the basic embedded-system process of sensing an environmental parameter, processing the sensor data using a microcontroller, making a decision, and controlling an output.

3. Objectives

- Understand Arduino Uno and basic embedded programming.
- Interface an LDR with an analog input.
- Read a sensor value using ADC.
- Control an LED through a digital output.
- Implement threshold-based automatic lighting.
- Design and test the system using software simulation.

4. Components

Component	Purpose
Arduino Uno	Main microcontroller
LDR / Photoresistor	Detects light intensity
LED	Represents the street light
220 Ω resistor	Limits LED current

5. Software Used

- Wokwi Online Simulator — circuit design and simulation.
- Arduino C/C++ — program development.

6. Circuit Connections

Component Pin	Arduino Connection
LDR VCC	5V
LDR GND	GND

LDR AO	A0
LDR DO	Not connected
Arduino D8	220 Ω resistor → LED
LED negative	GND

7. Working Principle

The LDR changes its electrical characteristics according to illumination. Its analog output is connected to A0. Arduino reads the value using `analogRead()`. The program compares the reading with a threshold of 500. If the reading is below 500, the system treats the condition as dark and turns the LED ON. Otherwise, it turns the LED OFF. The reading is printed on the Serial Monitor for debugging.

8. Program

```
int ldrPin = A0;
int ledPin = 8;

void setup() {
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int lightValue = analogRead(ldrPin);
  Serial.print("Light Value: ");
  Serial.println(lightValue);

  if (lightValue < 500) {
    digitalWrite(ledPin, HIGH);
  } else {
    digitalWrite(ledPin, LOW);
  }

  delay(500);
}
```

9. Algorithm

1. Start the program.
2. Configure the LED as an output and initialize Serial communication.
3. Read the LDR value from A0.
4. Display the value on the Serial Monitor.
5. Compare the value with 500.
6. If the value is below 500, turn the LED ON.
7. Otherwise, turn the LED OFF.
8. Repeat continuously.

10. Simulation and Testing

The circuit was implemented and tested in Wokwi. The simulated light level was varied and the LDR readings were observed through the Serial Monitor. The LED responded automatically according to the programmed threshold.

11. Result

The Automatic Street Light simulation was successfully implemented. The Arduino read the LDR sensor and automatically controlled the LED according to the detected light condition.

12. Applications

- Automatic street lighting.
- Energy-saving lighting control.
- Outdoor lighting automation.
- Smart-city lighting concepts.
- Embedded-systems learning.

13. Limitations

- This version is a software simulation and has not been physically implemented.
- The threshold may require calibration for real hardware.
- The current system controls one simulated light.

14. Future Scope

- Implement using physical Arduino and LDR hardware.
- Control multiple street lights.
- Upgrade to ESP32 for Wi-Fi/IoT connectivity.
- Add energy monitoring and remote status reporting.

15. Conclusion

The project demonstrates automatic lighting control using Arduino and an LDR. It provides practical understanding of analog sensing, ADC-based reading, threshold decisions, digital output control, and serial debugging. Wokwi provides a convenient environment to test the design before hardware implementation.

16. References

- Arduino documentation — Arduino Uno and programming concepts.
- Wokwi documentation — Arduino and component simulation.
- General technical references on LDRs and embedded systems.